Upcoming + Reminders

Find all links on the **course website:** cogs9.github.io/fa26

📝 **Assignments** (find links on course website):

- Reading Quiz 4 (DUE: WEDNESDAY)

All due at 11:59pm

- Assignment 1— Data Visualization (DUE: Wed week 7)
- Assignment 2 — p-hacking (DUE: Wed week 8)
- Reading Quiz 5 (DUE: Wed week 9)
- Assignment 3 — Machine learning (DUE: Wed week 10)

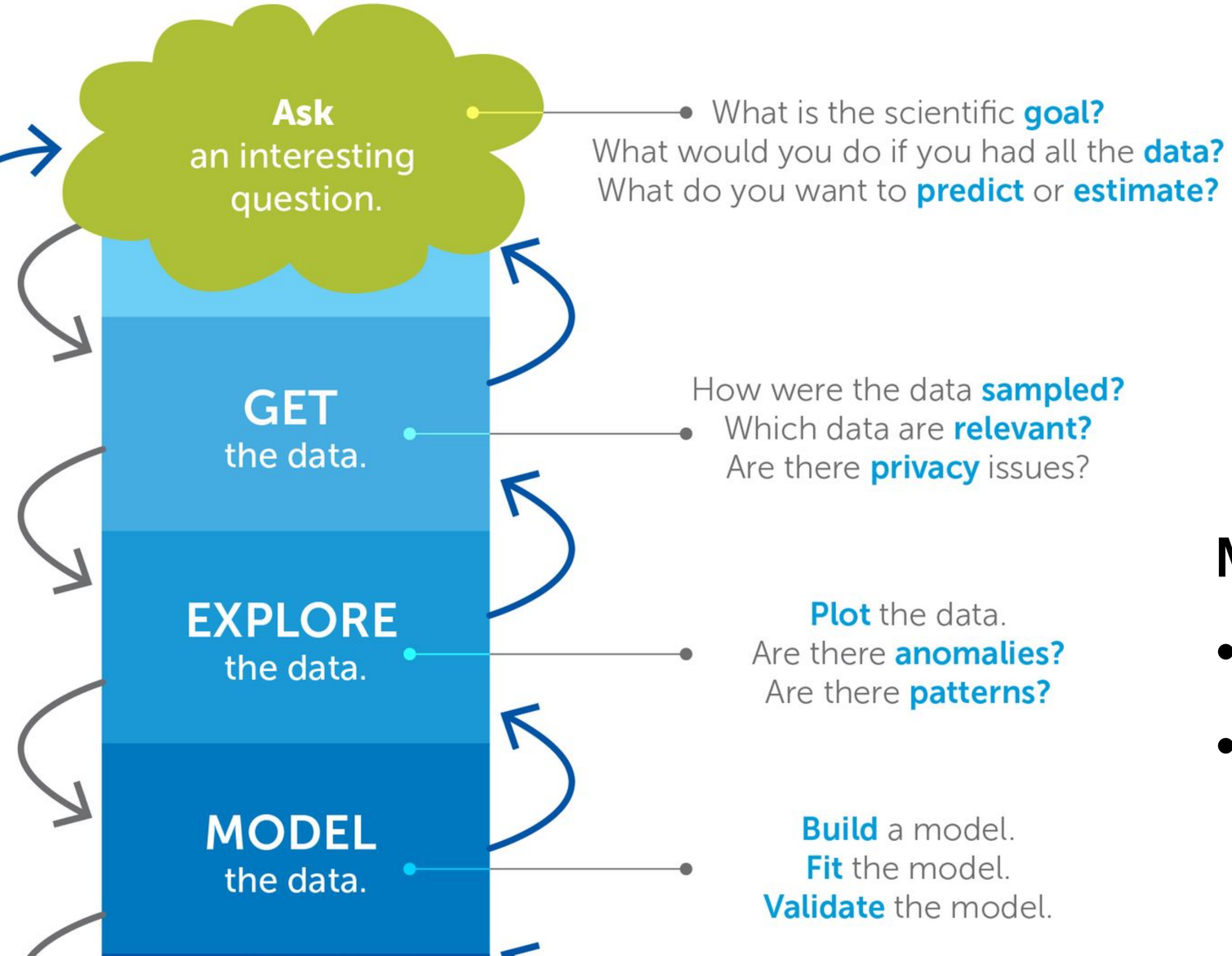
Midterm 2 - Monday Week 9 (November 23)

COGS 9 | Week 6

Machine Learning: Supervised Learning & Evaluation

Brendan Tomoschuk

Today: Machine Learning



Modeling the data in order to

- make predictions on new data
- make decisions based on those predictions

Machine Learning

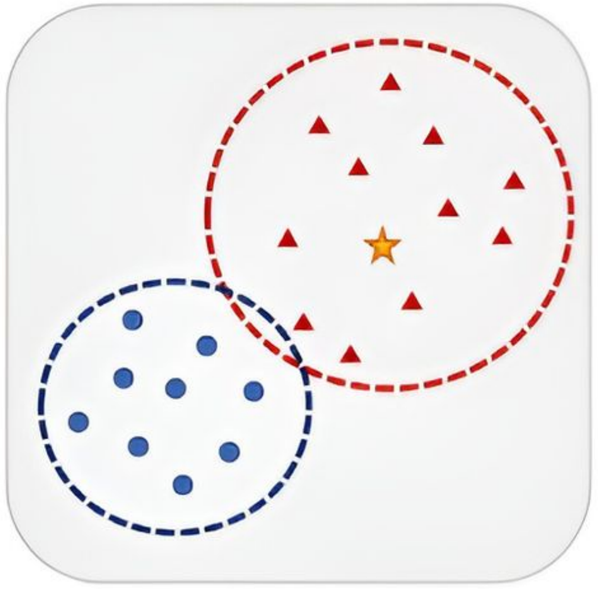
Learning from data to predict or generalize to unseen data

Learning from trial-and-error interactions with the environment

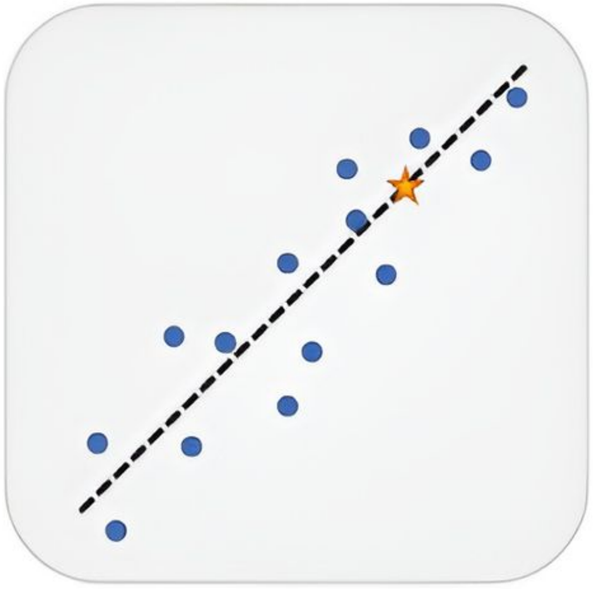
Supervised Learning

Model training with labelled data

Classification



Regression

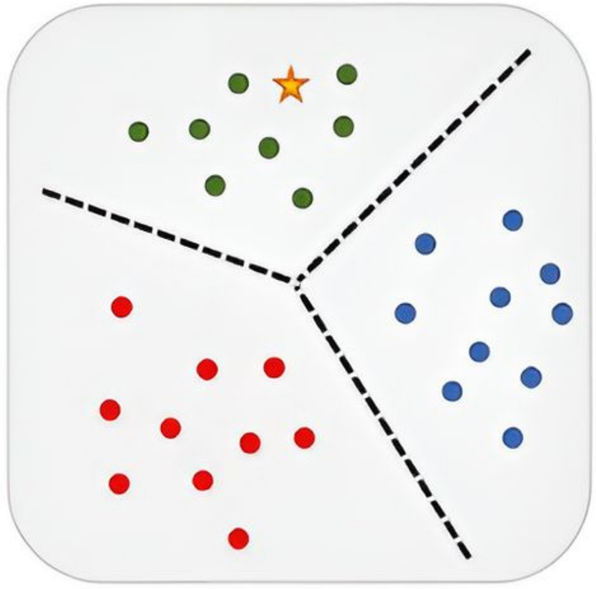


Unsupervised Learning

Model training with unlabelled data

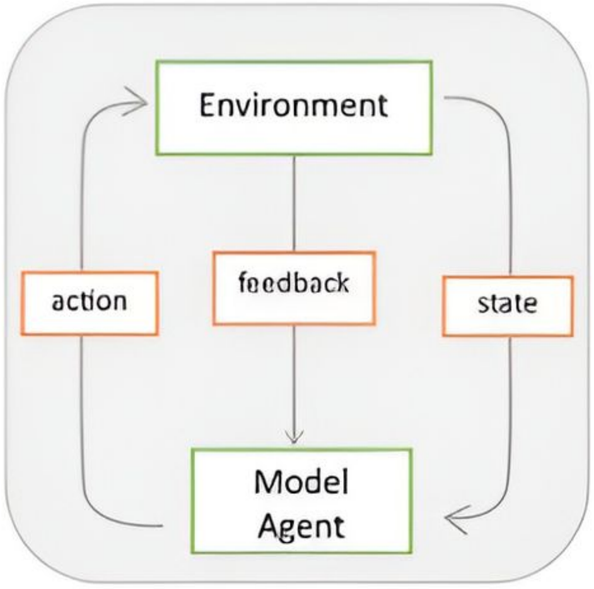
Discovering patterns in data

Clustering



Reinforcement Learning

Model take actions in the environment then received state updates and feedbacks



Evaluation Learning Goals

1. Finish up supervised learning models
   - *simple & multiple linear regression*
   - *non-linear regression*
   - *logistic regression*
   - *decision trees*
   - *neural networks*
2. Describe how supervised learning models are evaluated
3. Explain the importance of generalization and splitting data into training and test sets
4. Identify and diagnose overfitting vs. underfitting

Classification

Predicting qualitative Y

Classification (Predicting qualitative Y)

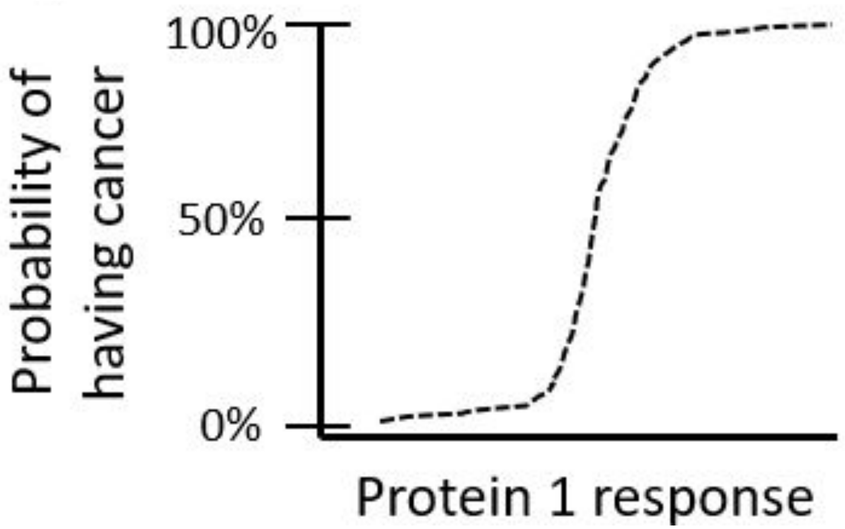
Classification models predict the **probability** of a data point belonging to a category

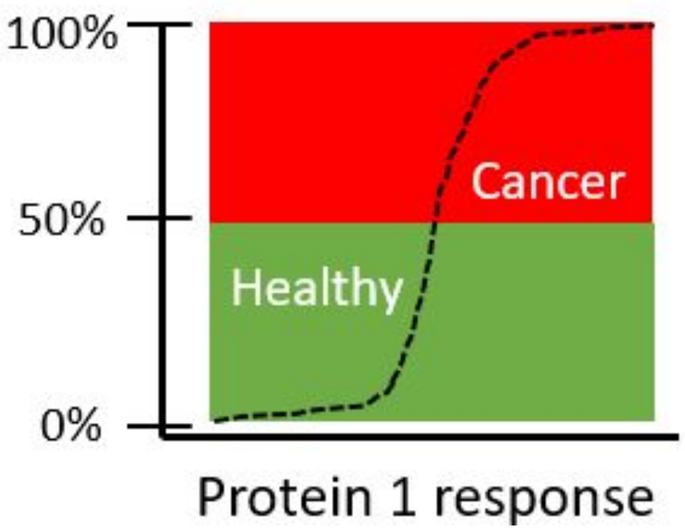
probability (that Y is in category k, given X

$$P(Y = k \mid X) = f(X)$$

output, response

category or class

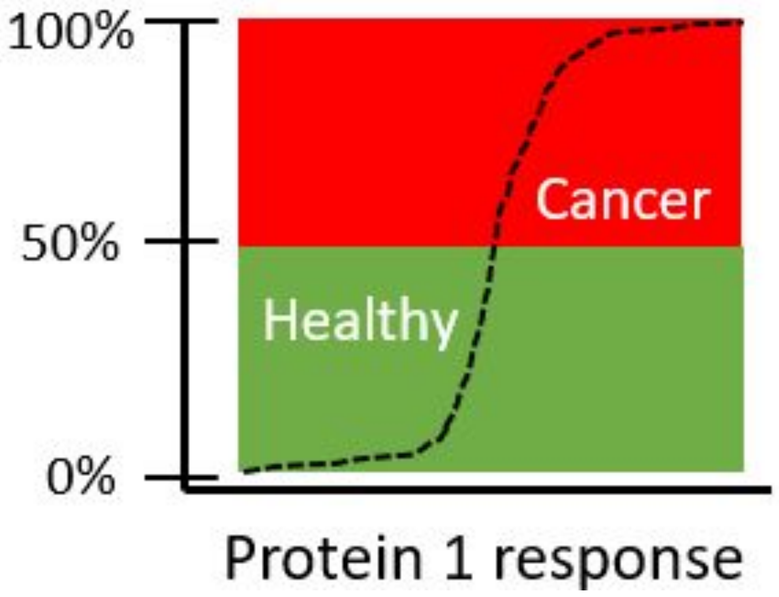


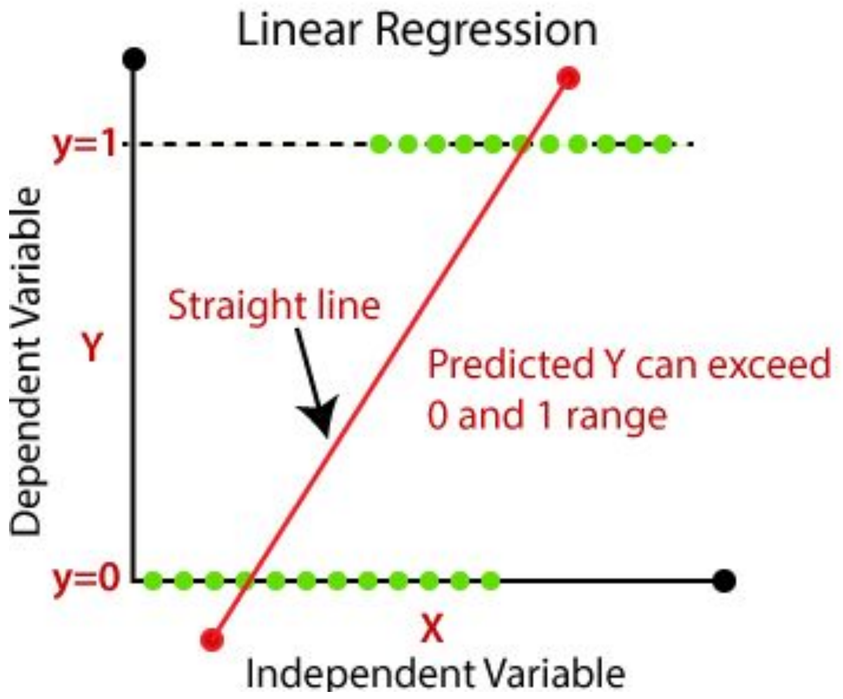


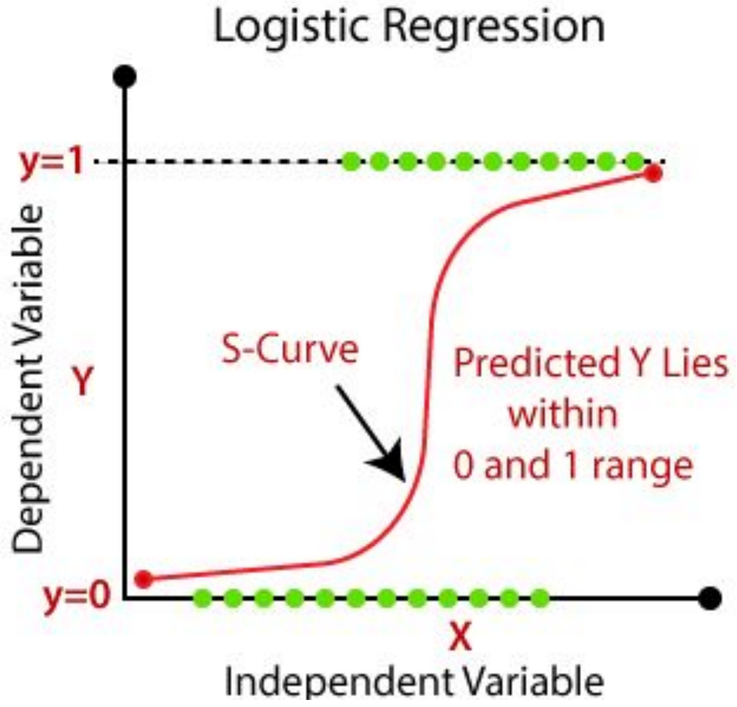
Logistic Regression

Logistic regression: models the relationship between predictors and the probability of an outcomes

- f is a *sigmoid function* (S-shaped curve) and outputs a probability
- Linear regression **CANNOT** accommodate **categorical** responses







Logistic Regression

<u>Simple</u> logistic regression model (one X), binary classification (outcome is 0 or 1)

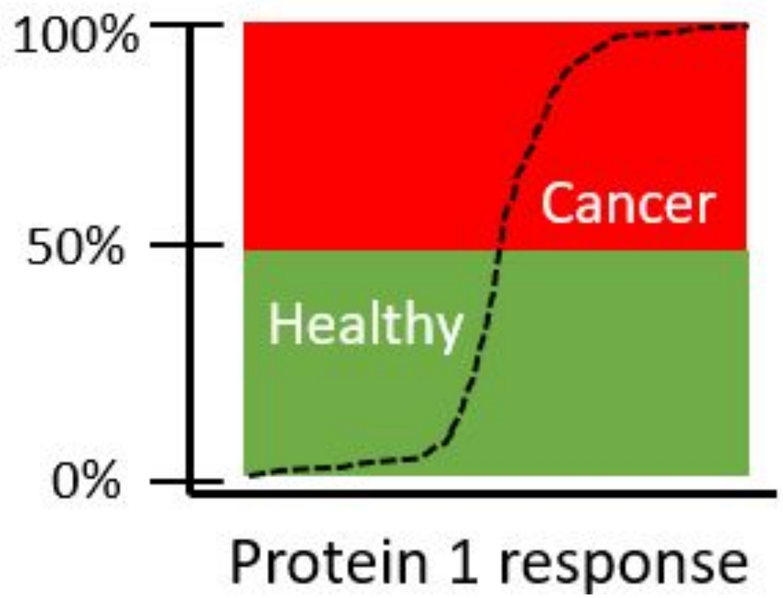
unknown *parameters*

probability (that Y is in category 1)

$$P(Y = 1 \mid X) = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

output, respons

predictors, features



Classification rule: $Y = \begin{cases} 1, & \text{if } P(Y = 1 \mid X) > 0.5 \\ 0, & \text{otherwise.} \end{cases}$

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course code:
WSTAPX

You want to predict whether or not you will get an A in COGS 9 this quarter. Which model should you use?

(A) Multiple linear regression

(B) Simple linear regression

(C) Polynomial regression

(D) Logistic regression

(E) Any of the above could work

Webclicker

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You want to predict whether or not you will get an A in COGS 9 this quarter. Which model should you use?

(A) Multiple linear regression

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Supervised Learning Models

Regression: predicting quantitative responses

Models:

- Linear regression (simple & multiple)
- Non-linear regression
- Ridge & lasso regression

Both regression & classification:

- Decision Trees
- Random Forests & Boosting
- K-Nearest Neighbors
- Neural Networks

Classification: predicting qualitative (categorical) responses

Models:

- Logistic regression
- Linear Discriminant Analysis (LDA)
- Quadratic Discriminant Analysis (QDA)
- Naive Bayes
- Support Vector Machine (SVM)

Takeaway: Your choice of model depends on the *outcome* you are predicting as well as the *relationship* between the input and output!

Supervised Learning Models

Supervised Learning: Predicting Y from X

Regression: Predicting quantitative Y

- Linear regression (simple & multiple)
- Non-linear regression

Classification: Predicting qualitative (categorical) Y

- Logistic regression

Both regression & classification:

- Decision trees
- Neural networks

Takeaway: Your choice of model depends on the *outcome* you are predicting as well as the *relationship* between the input and output!

Both Regression & Classification

Models that can predict

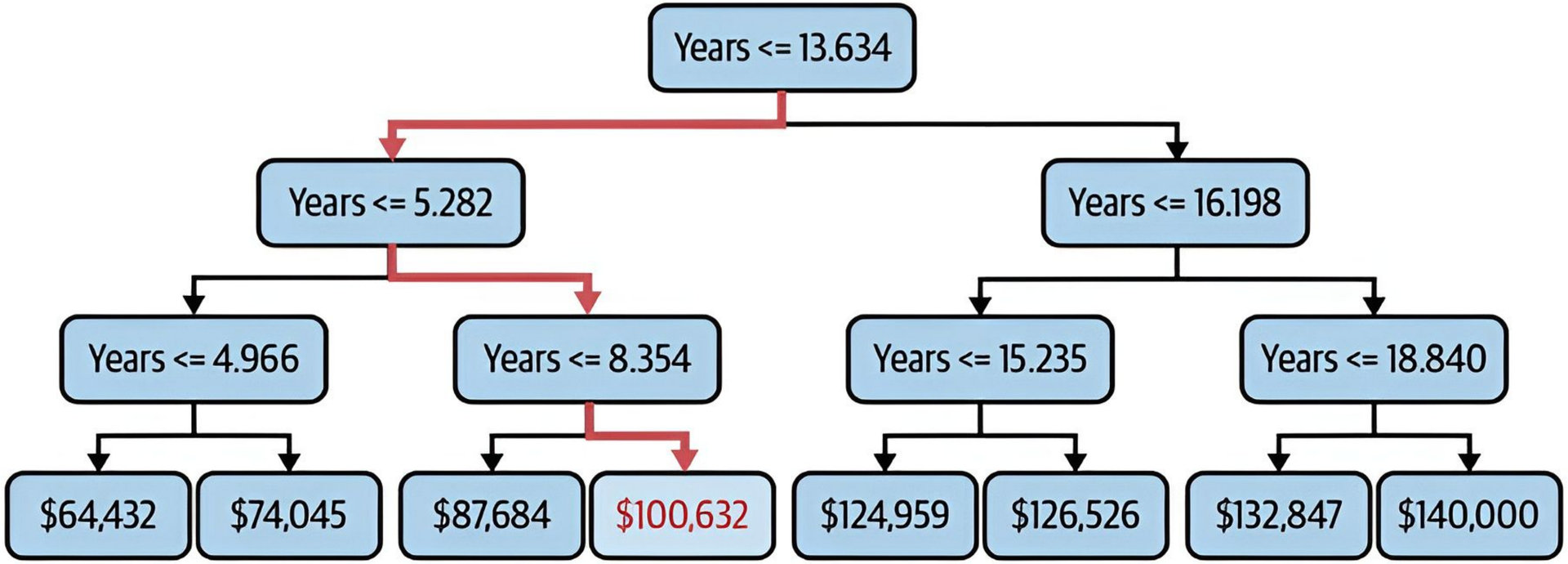
both quantitative and qualitative Y

Decision Trees

Decision Trees: a model that makes predictions by **splitting** the predictors into a series of "branches" that can be summarized in a "tree"-like visual

- Good for modeling ***non-linear*** relationships between X and Y

Regression Decision Tree: Predicting salary from years of experience



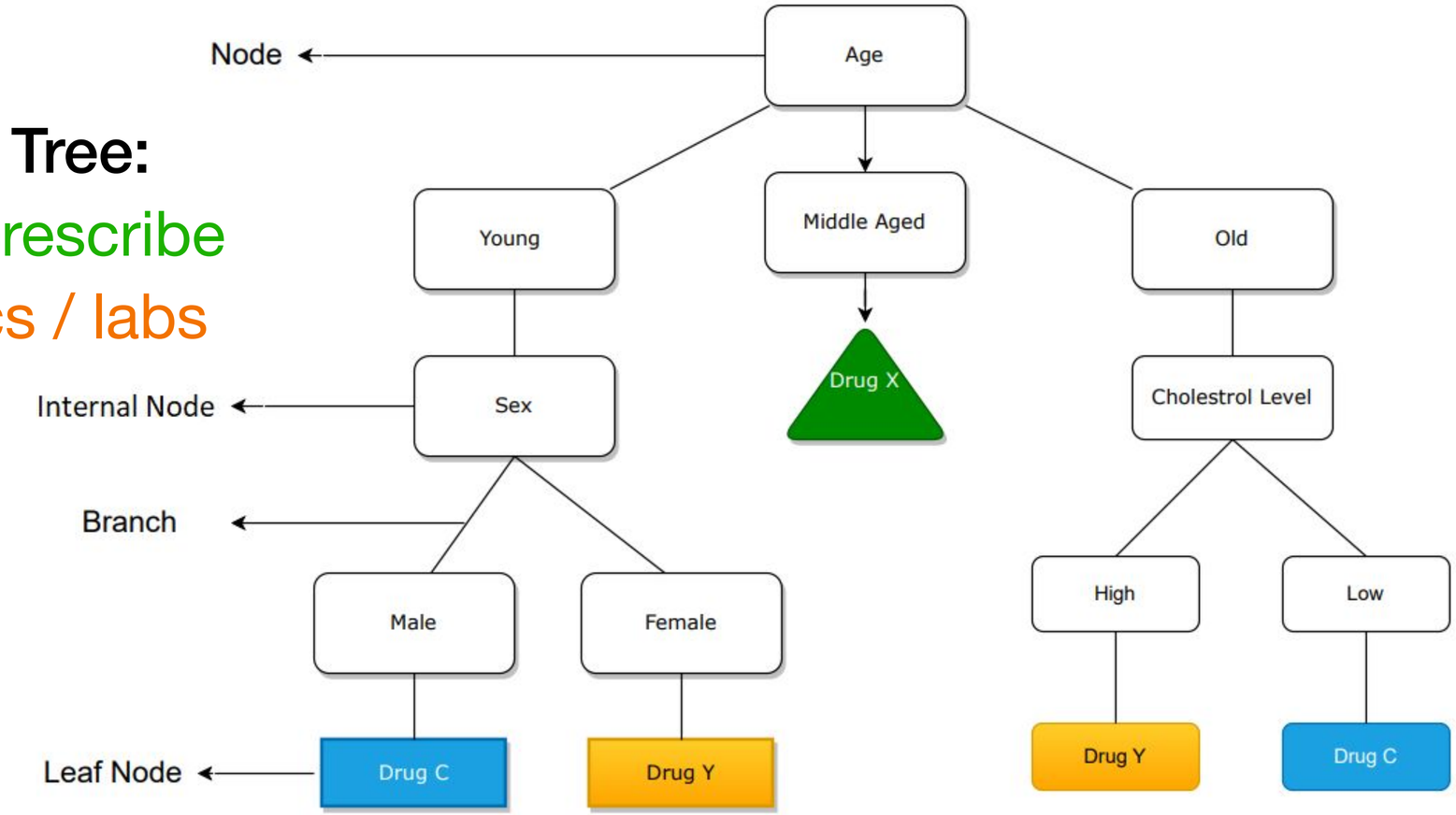
Decision Trees

Decision Trees: a model that makes predictions by **splitting** the predictors into a series of “branches” that can be summarized in a “tree”-like visual

- Good for modeling ***non-linear*** relationships between X and Y

Classification Decision Tree:

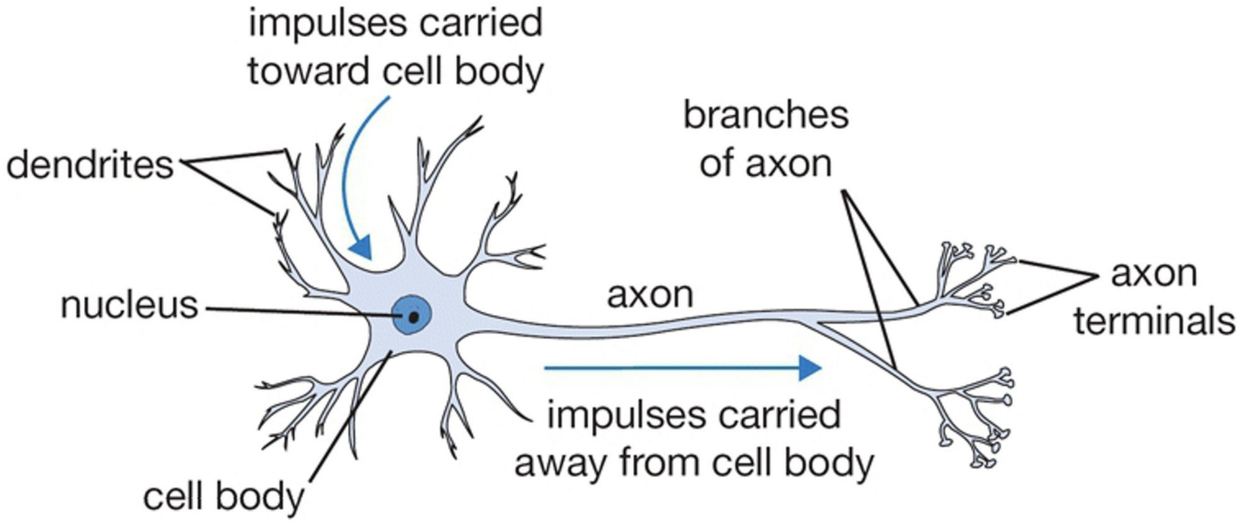
Predicting which drug to prescribe from patient demographics / labs

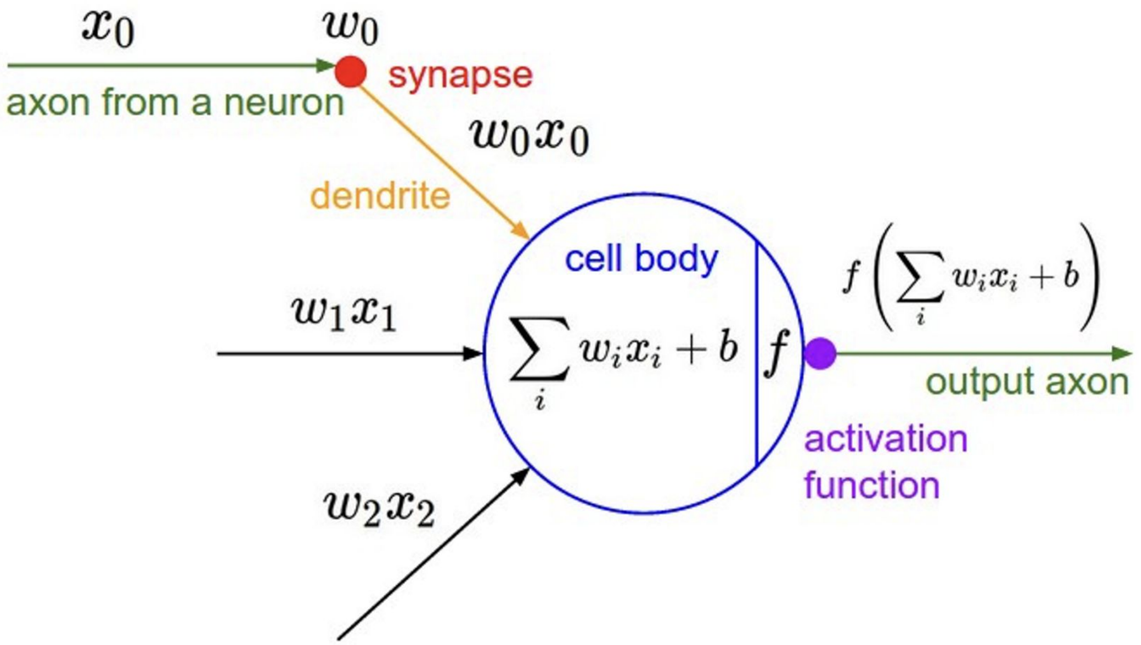


Neural Networks

Neural networks are computational models inspired by the structure and function of neurons in the brain

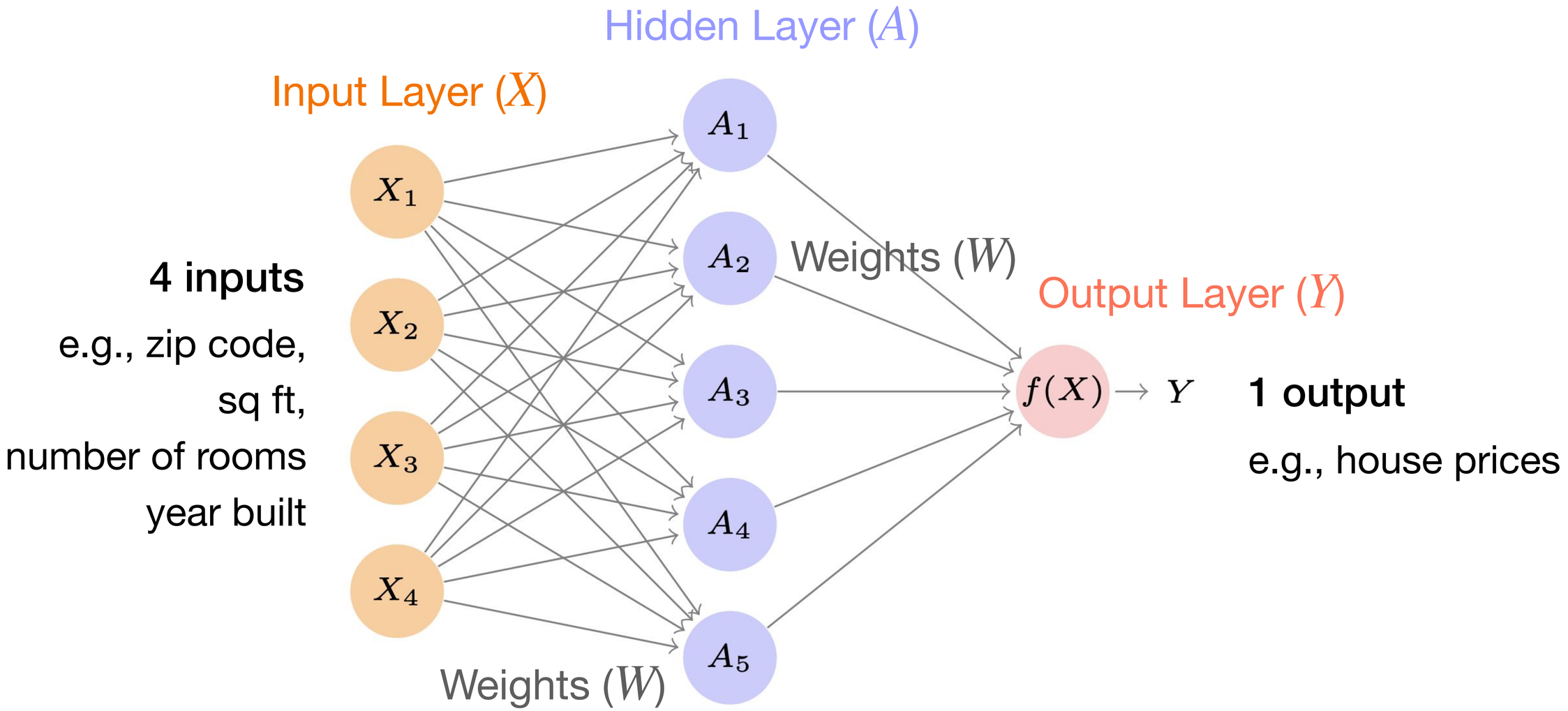
- They learn to recognize **patterns** in data by adjusting their parameters to capture complex, ***non-linear*** relationships between X and Y
- **Applications:** Image/speech/face recognition, LLMs (transformers), self-driving cars, medical diagnosis





Neural Networks

Goal: estimate the weights (W) that relate the inputs (X) and outputs (Y)

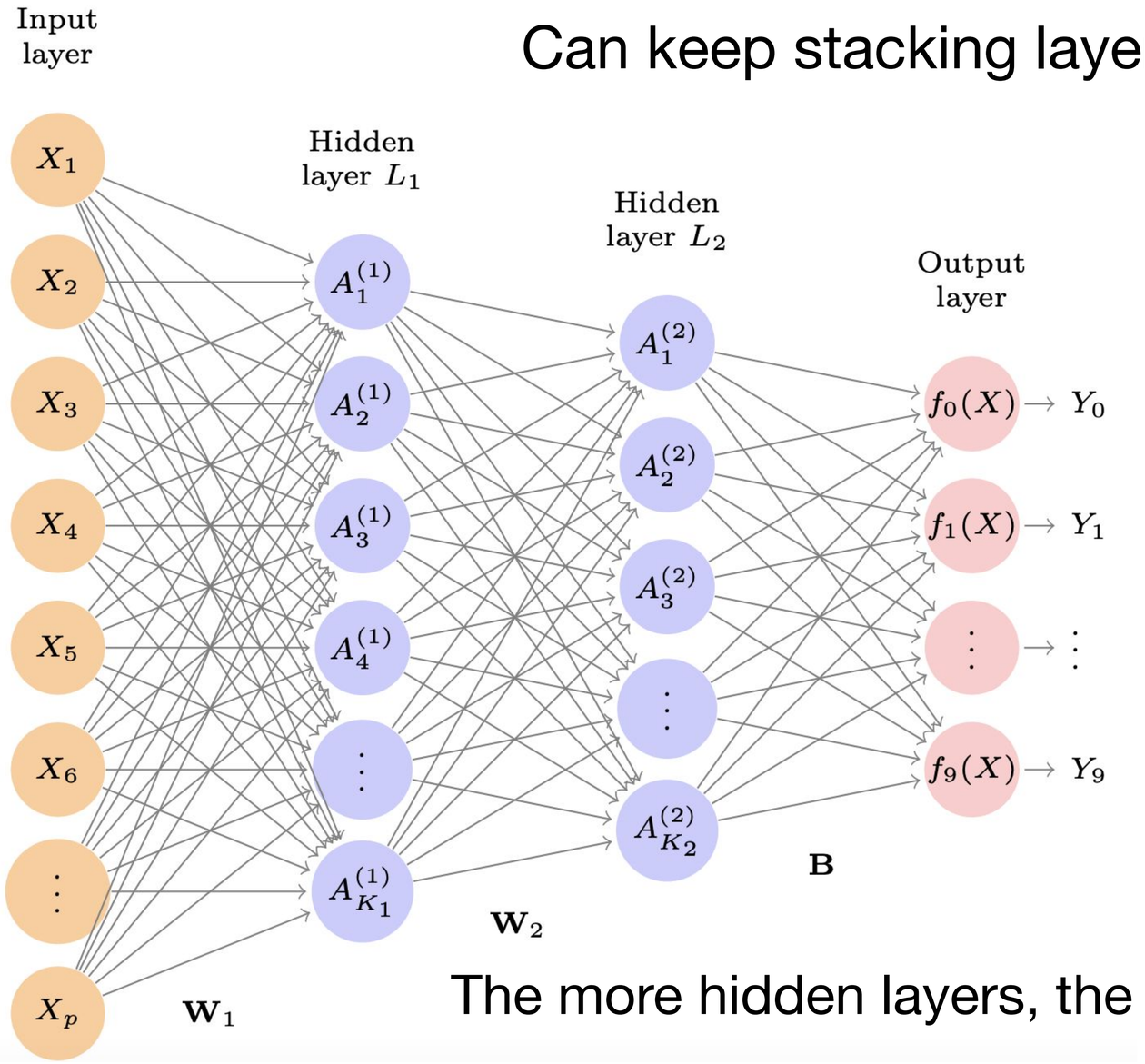


Neural Networks

"Feed-forward neural network"

Can keep stacking layers on top of each other!

p inputs

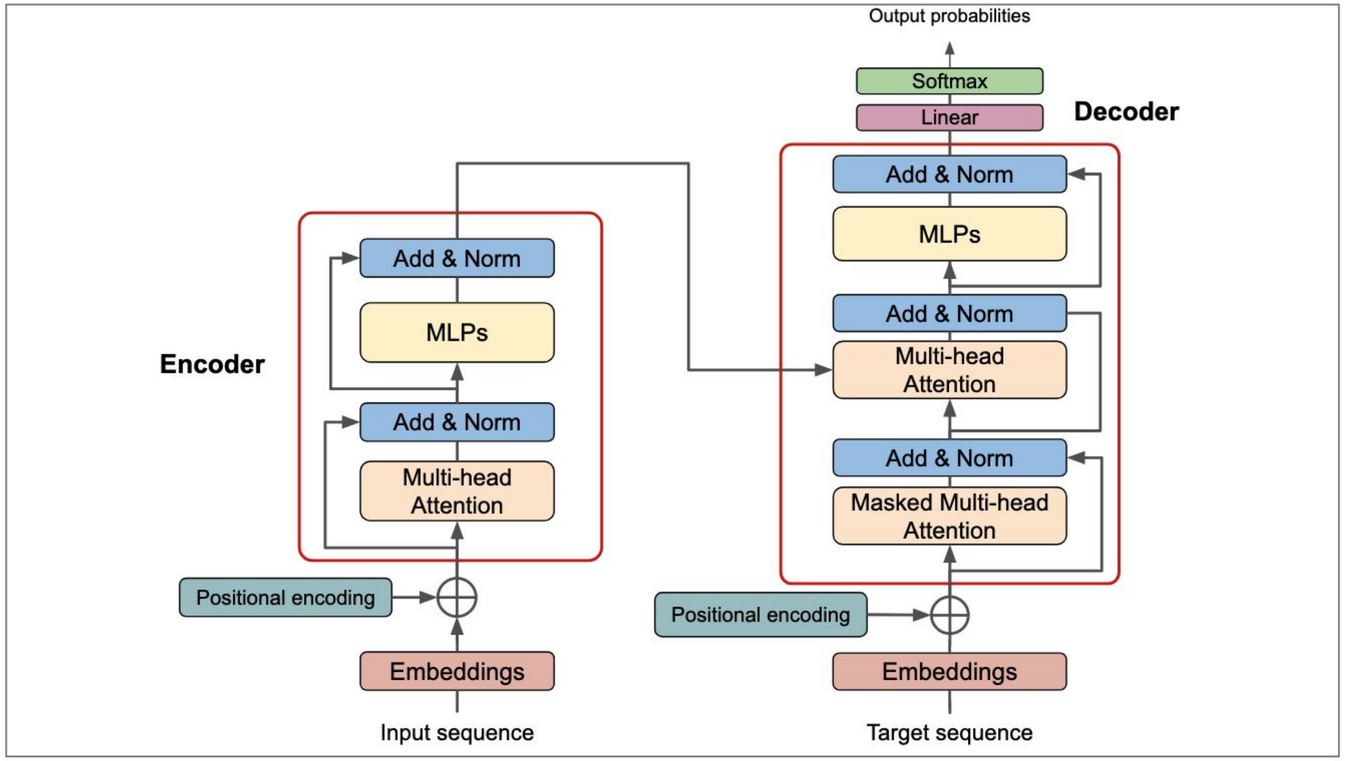


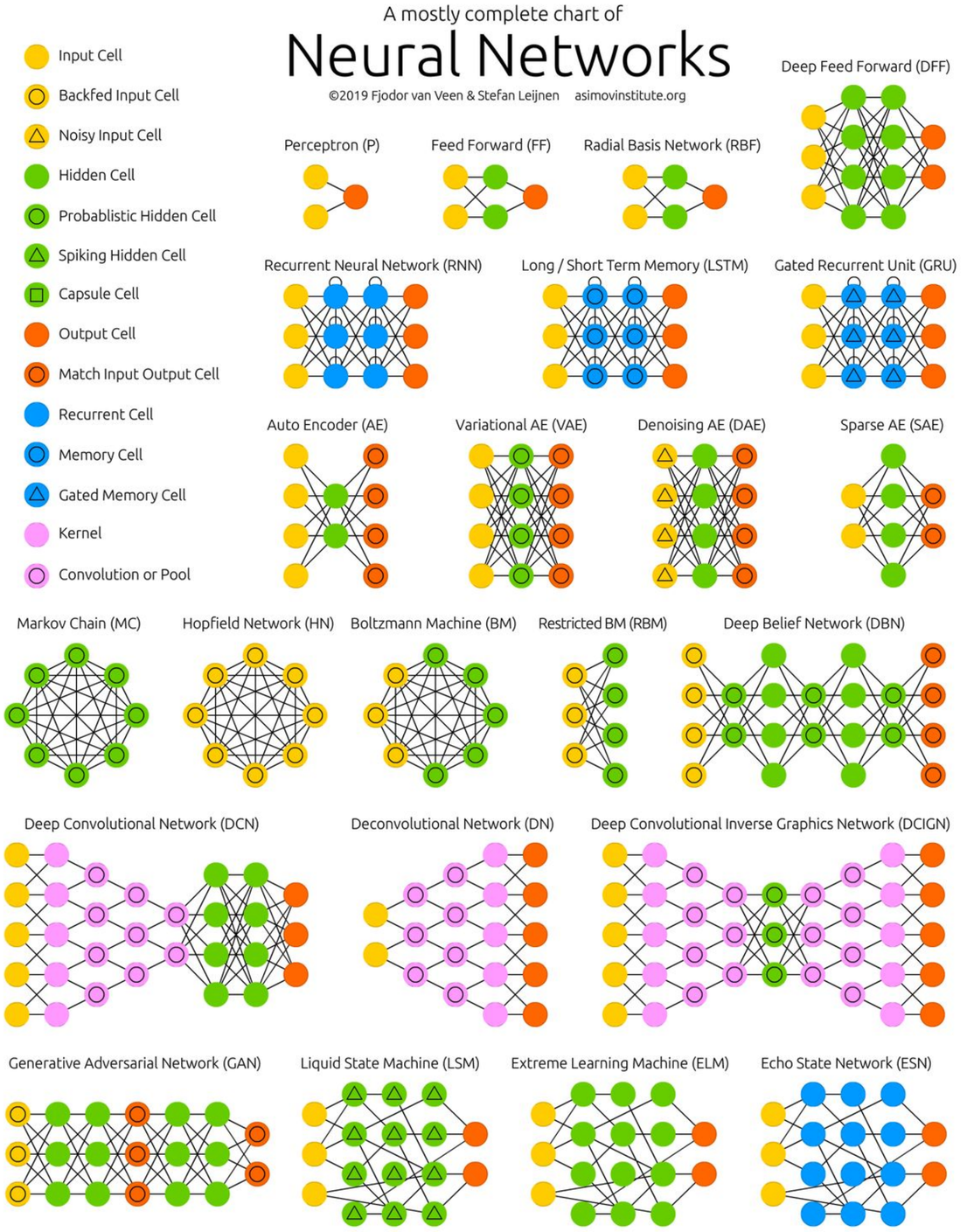
9 outputs

The more hidden layers, the "deeper" the neural network

There are many, many types of neural networks…

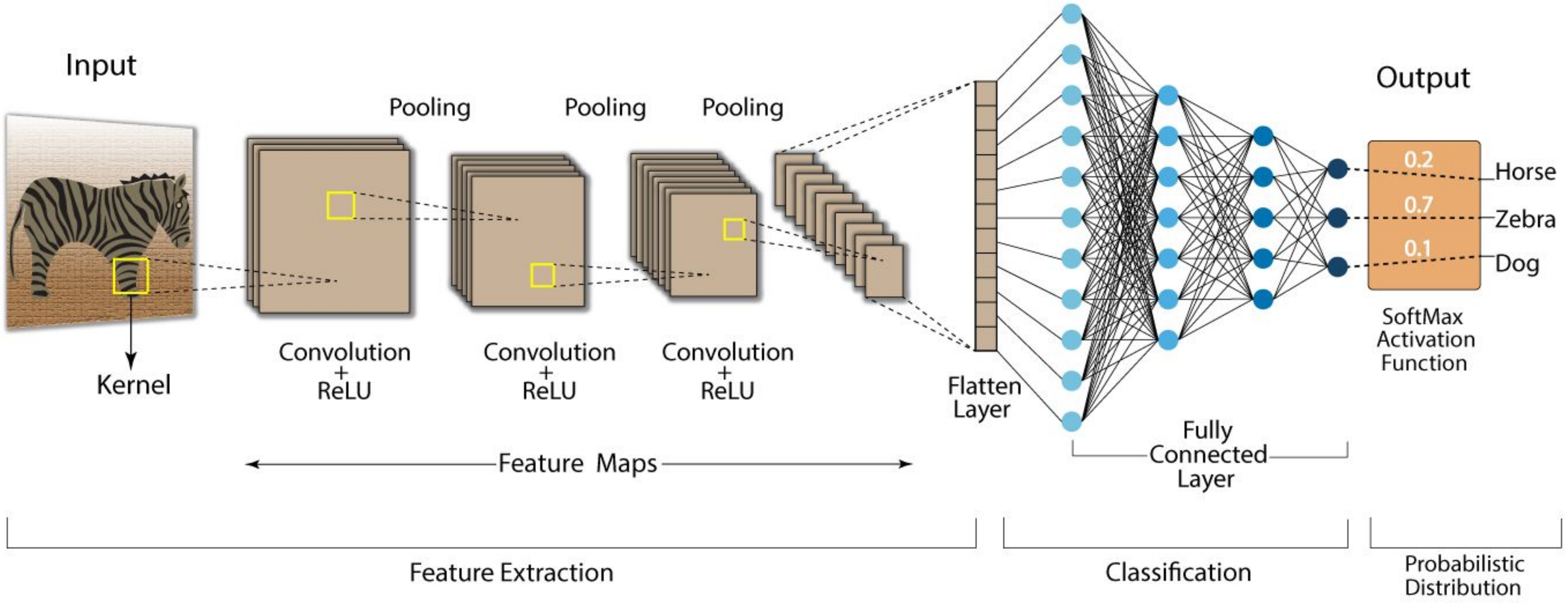
Large language models (LLMs) like ChatGPT are a special type of neural network called **transformers:**



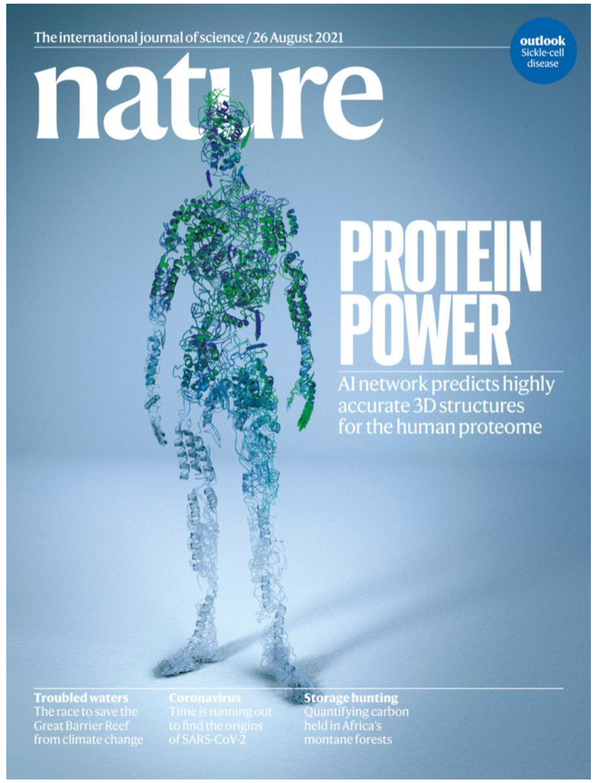


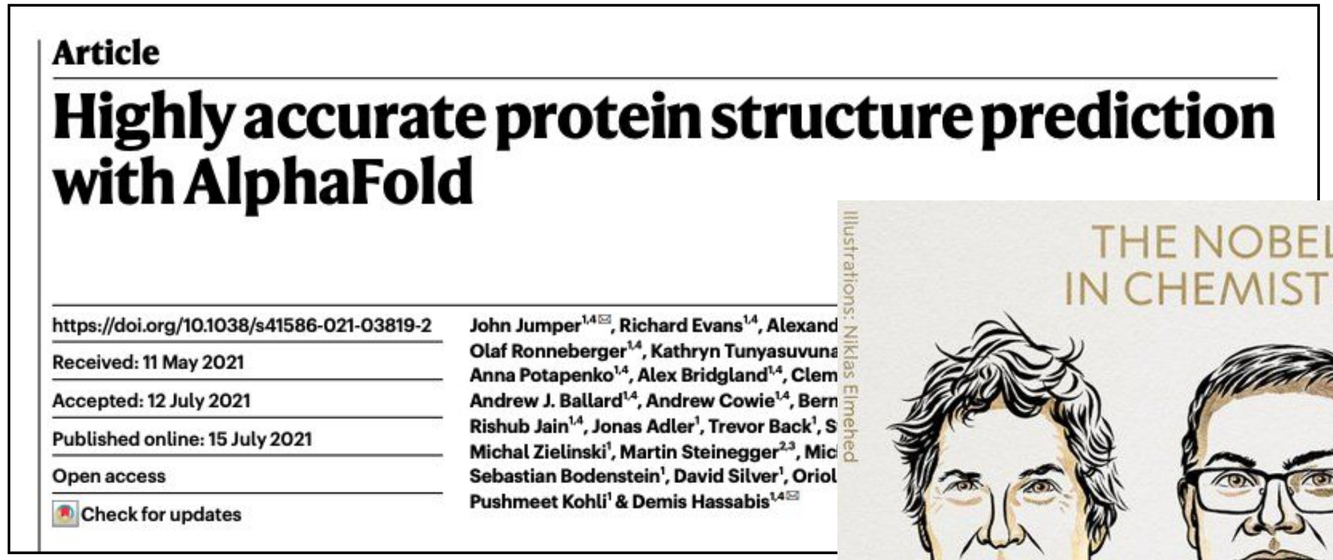
There are many, many types of neural networks…

Convolutional neural networks (CNNs) can be trained to classify objects and images:



AI wins the Nobel Prize with deep neural nets

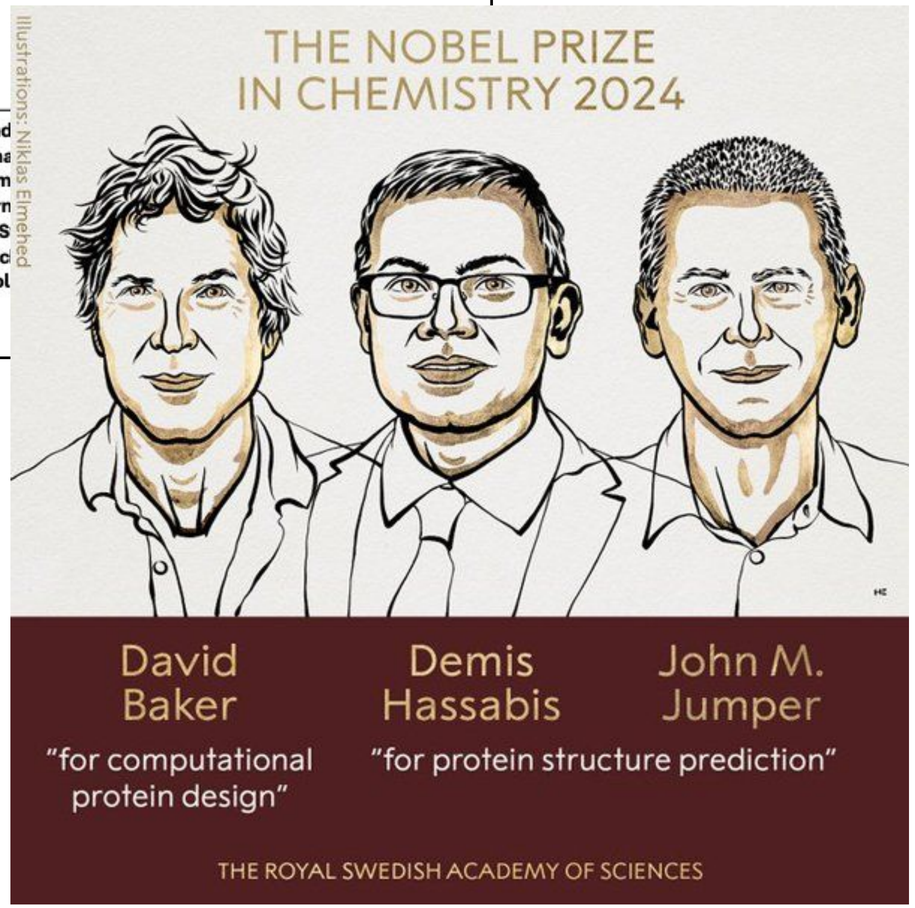



Article

Highly accurate protein structure prediction with AlphaFold

https://doi.org/10.1038/s41586-021-03819-2
Received: 11 May 2021
Accepted: 12 July 2021
Published online: 15 July 2021
Open access
Check for updates

John Jumper[1,4]✉, Richard Evans[1,4], Alexand
Olaf Ronneberger[1,4], Kathryn Tunyasuvuna
Anna Potapenko[1,4], Alex Bridgland[1,4], Clem
Andrew J. Ballard[1,4], Andrew Cowie[1,4], Bern
Rishub Jain[1,4], Jonas Adler[1], Trevor Back[1], S
Michal Zielinski[1], Martin Steinegger[2,3], Mic
Sebastian Bodenstein[1], David Silver[1], Oriol
Pushmeet Kohli[1] & Demis Hassabis[1,4]✉



- AlphaFold showed it could accurately predict protein structures in **minutes** (used to take human researchers multiple *years* to solve just one!)
- AI "won" the Nobel Prize in Chemistry in 2024!

Why aren’t all data scientists just using neural networks?

Neural networks

- Often requires **more** data
- **Low** interpretability
- Requires **high** compute
- Performs well for **unstructured** data
- **Less** reliance on domain expertise

Other models (regression, trees etc)

- Often requires **less** data
- **High** interpretability
- Requires **less** compute
- Performs well for **structured** data
- **More** reliance on domain expertise

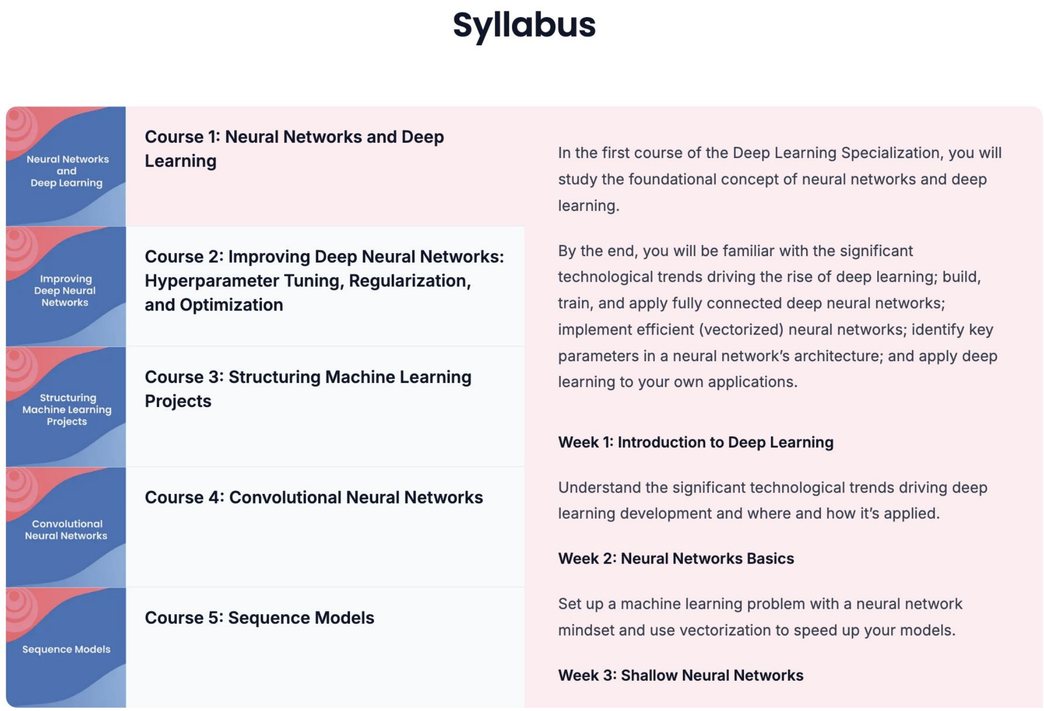
To continue learning about neural networks…

At UCSD

- CSE 151B: Deep Learning
- COGS 181: Neural Networks and Deep Learning
- ECE 176: Intro to Deep Learning & Applications

Online:

- Coursera Deep Learning Specialization
- Tensorflow Playground



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course code:
ORXHTC

You want to predict who will win the NBA Finals from their previous game statistics. You suspect that the relationship between the input and output is non-linear.

Which model(s) could you use? *Select all…*

(A) Multiple linear regression

(B) Decision tree

(C) Neural network

(D) Logistic regression

(E) B, C, & D

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Takeaway: Your choice of model depends on the *outcome* you are predicting as well as the *relationship* between the input and output!

Model Evaluation

For supervised learning models ONLY

How do we know if our model is good?

With a neighbor (~2 mins):

If a model is 100% accurate in predicting **whether someone has cancer or not** on a small dataset N=100, is this a good or bad model? *Why?*

How do we know if our model is good?

- The goal of supervised learning is to be able to make predictions on **new, unseen data = *"generalization"***
- To do that, we **split** our dataset: use part of it to *train* the model (learn $\hat{f}$), and part of it to *test* the model (the "unseen" data)

unknown function to learn

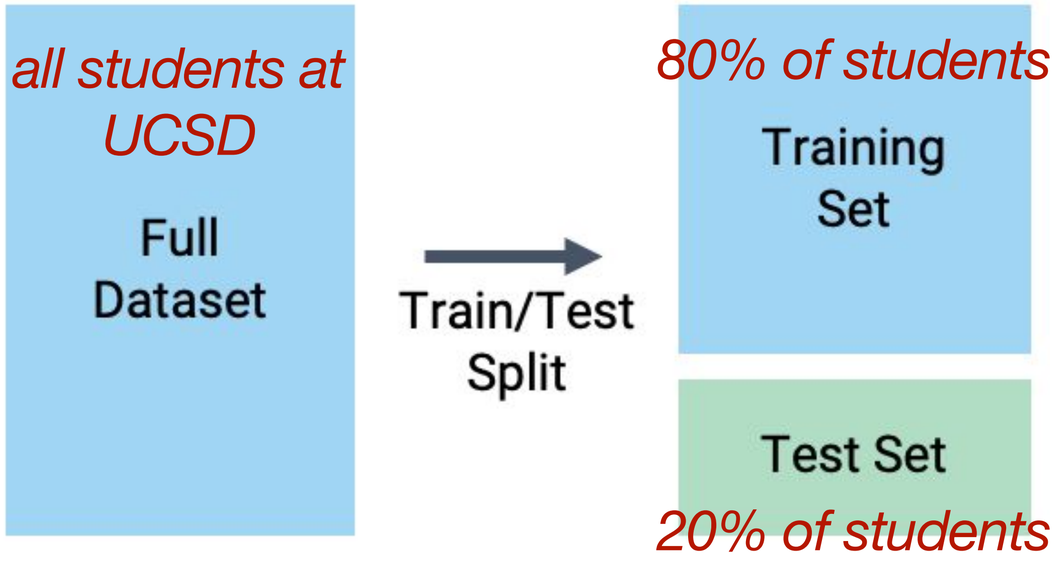
Regression $Y = f(X)$

output predictors

Classification

$$P(Y = k \mid X) = f(X)$$

Example: Predicting whether someone will get into UCSD or not from HS GPA



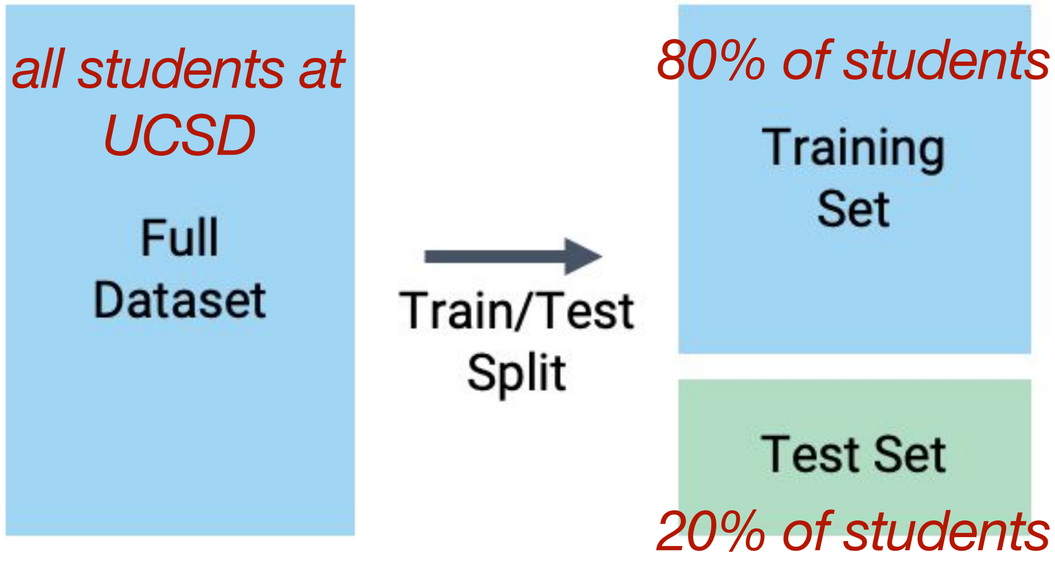
How do we know if our model is good?

Model evaluation: measuring how well a supervised learning model *generalizes*, or how it performs on data it hasn't seen before *(the "test" data)*

Example:

1. Train the model on 80% of student data ("training data"): { x_i , y_i } pairs
2. Then, take the x_i's (HS GAP) of the other 20% of data ("test data") to predict y_i (whether they will get into UCSD or not)

Example: Predicting whether someone will get into UCSD or not from HS GPA



Classification

$$P(Y = k \mid X) = f(X)$$

How do we know if our model is good?

Model evaluation metric: quantifies how well the model's predictions match the actual data

- For **classification** (qualitative Ys): the evaluation metric is **accuracy rate**
- *For a given point x_i, how often did the model predict $\hat{f}(x_i) = y_i$? (How often did it accurately predict that a "test" student will get into UCSD or not?)*

Actual category

$$\text{acc. rate} = \frac{1}{n}\sum_{i=1}^{n} I\left(y_i = \hat{f}(x_i)\right)$$

Predicted category

$I = 1$ if $y_i = \hat{f}(x_i)$

Confusion matrix

Predicted Y \ Actual Y	Reject (0)	Admit (1)
Reject (0)	True -	False -
Admit (1)	False +	True +

How do we know if our model is good?

Model evaluation metric: quantifies how well the model's predictions match the actual data

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Actual category

$$\text{acc. rate} = \frac{1}{n}\sum_{i=1}^{n} I\left(y_i = \hat{f}(x_i)\right)$$

Predicted category

$I = 1$ if $y_i = \hat{f}(x_i)$

Confusion matrix

Predicted Y \ Actual Y	Reject (0)	Admit (1)
Reject (0)	7000	300
Admit (1)	400	2300

How do we know if our model is good?

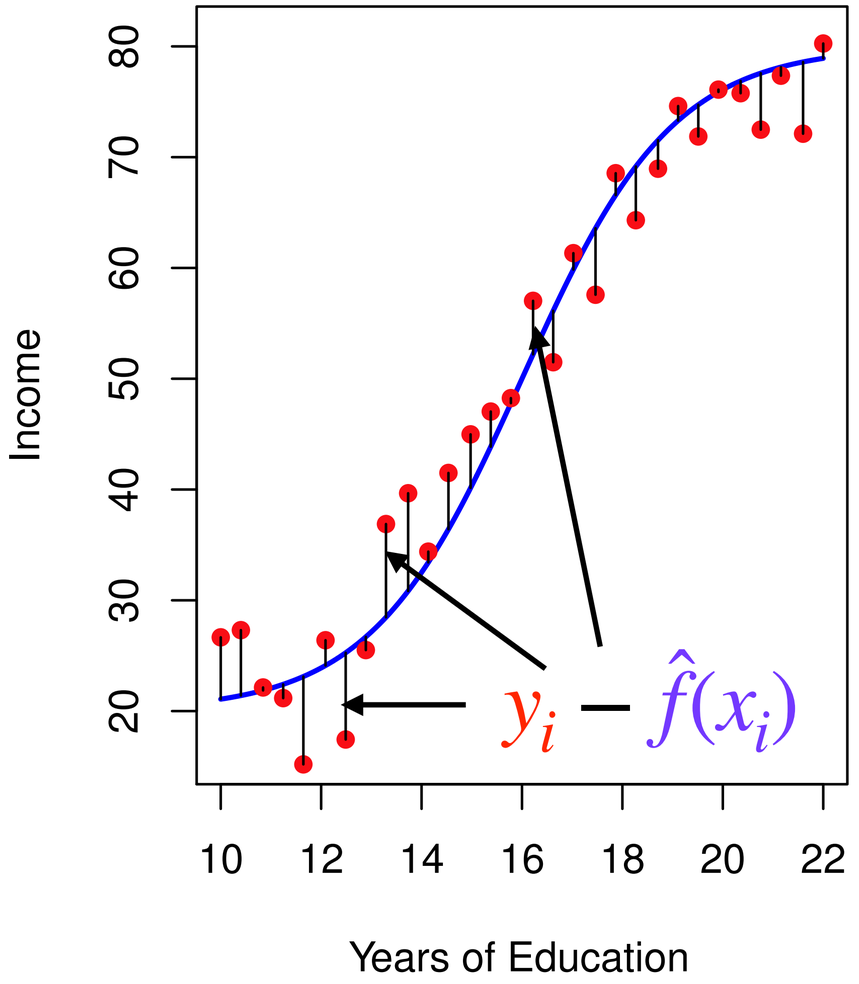
Model evaluation metric: quantifies how well the model's predictions match the actual data

- For **regression** (quantitative Ys): a common evaluation metric is **mean squared error (MSE)**
- *For a given point x_i, how close is $\hat{f}(x_i)$ to y_i?*

Actual exam score

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$$

Predicted exam score

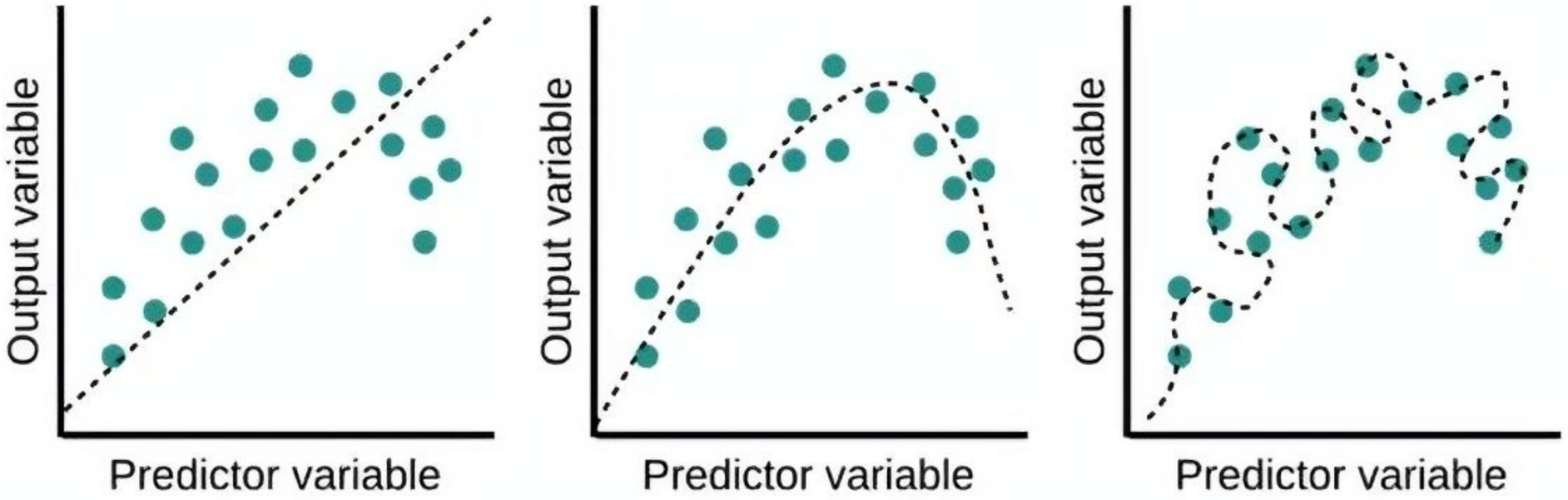


Discussion (~2-3 minutes)

Imagine that the teal data points are your **training data.**

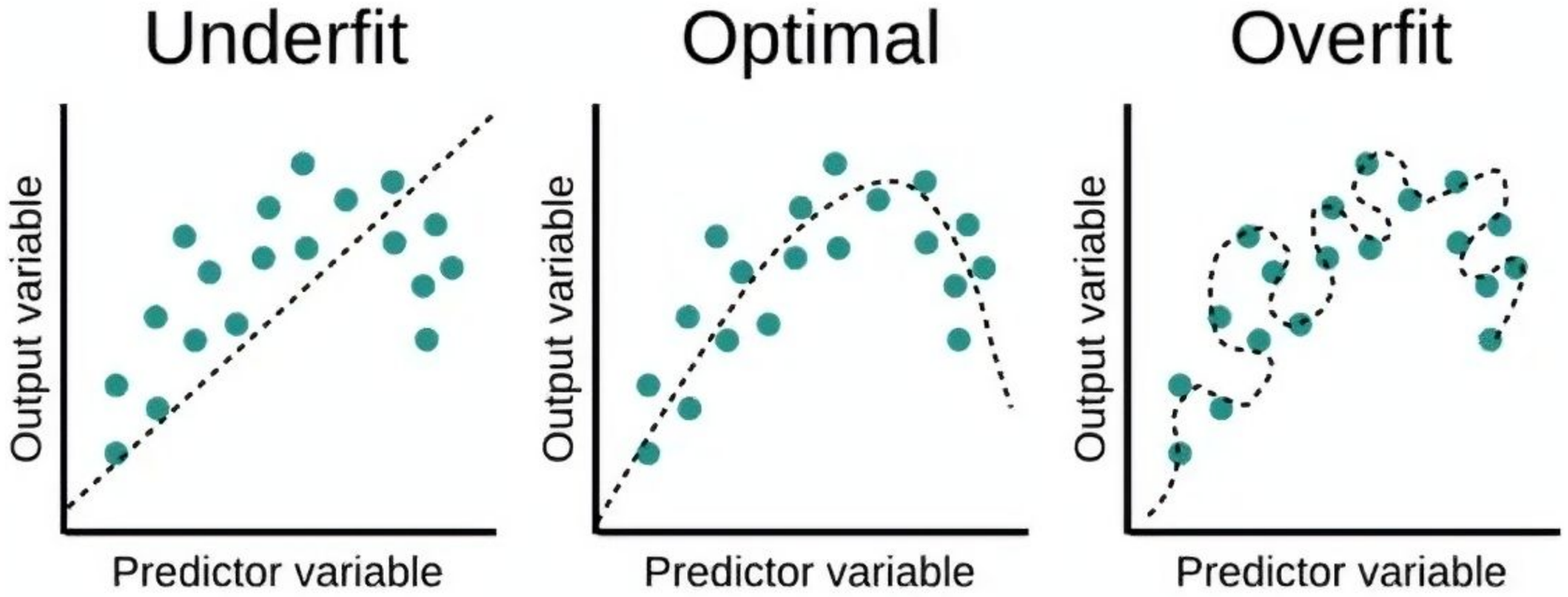
- Which of the following models (dotted lines/curves) do you think would **generalize the best on unseen test data**?

- Which has the highest MSE? lowest MSE?



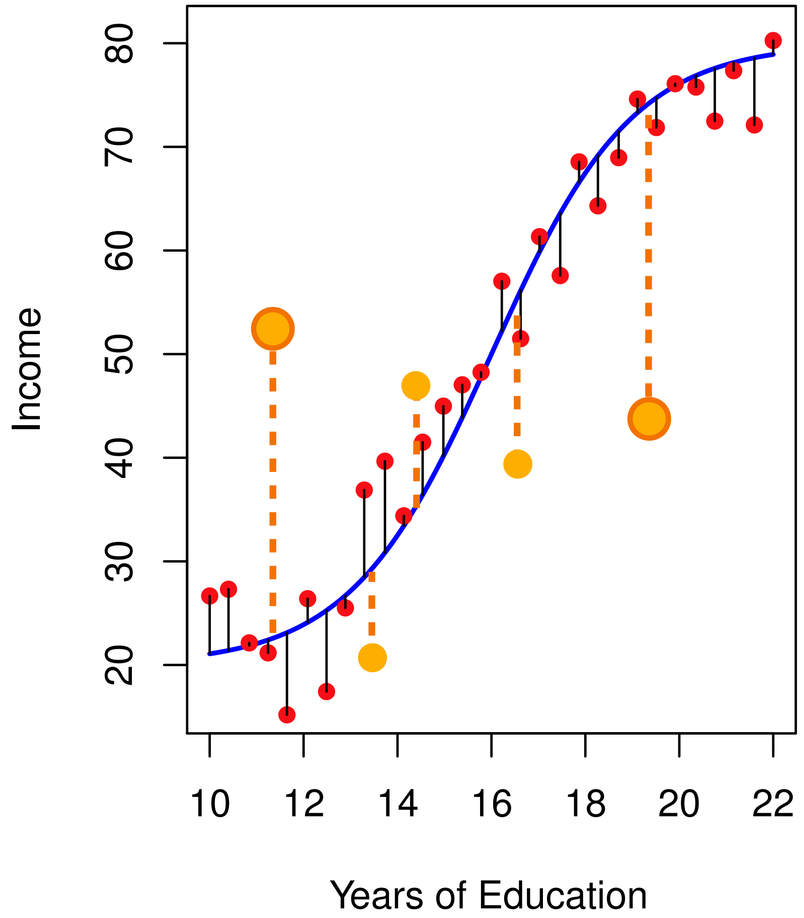
Overfitting vs. underfitting

- **Underfitting:** the model is *too simple* for the training data, performs poorly on both training and test data
- **Overfitting:** the model fits the training data *too well* (0 error, perfectly accurate) but would fit poorly on any new, test data
- *How do we make sure our model is not under or overfitting*?



Comparing train and test error for generalization

- You can measure how well the model ***generalizes*** by comparing train and test error
  - **Example:** Predicting someone's income from years of education



$\hat{f}$ estimated from training data

unseen test data

test mean squared error (MSE)

test data with large test MSE

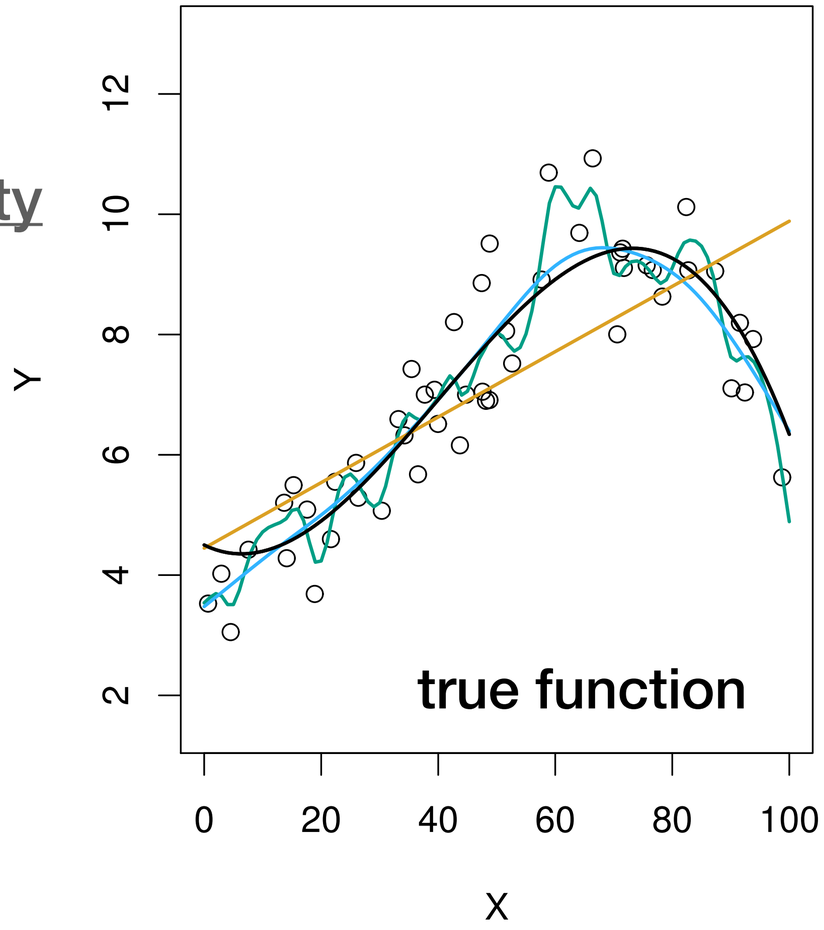
Comparing train and test error for generalization

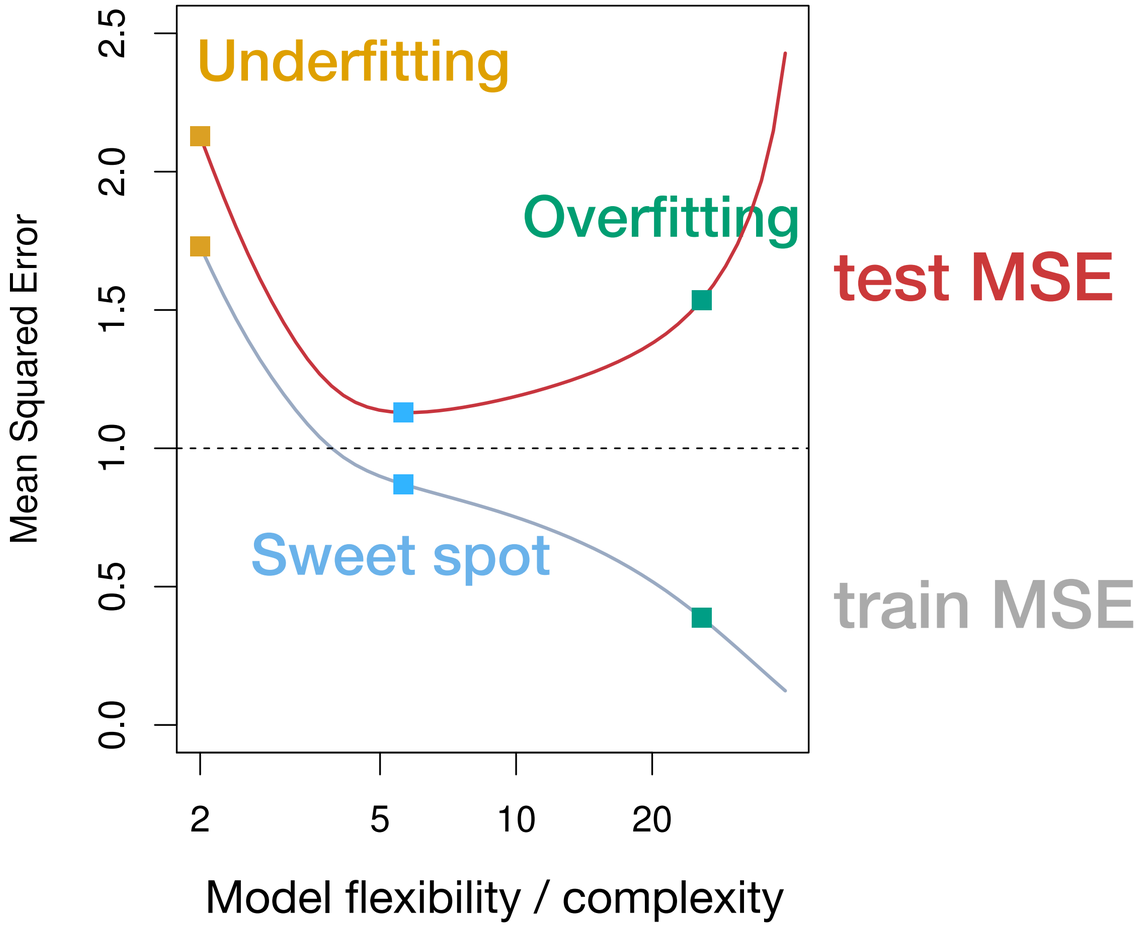
Model flexibility

Least flexible (linear)

Moderate

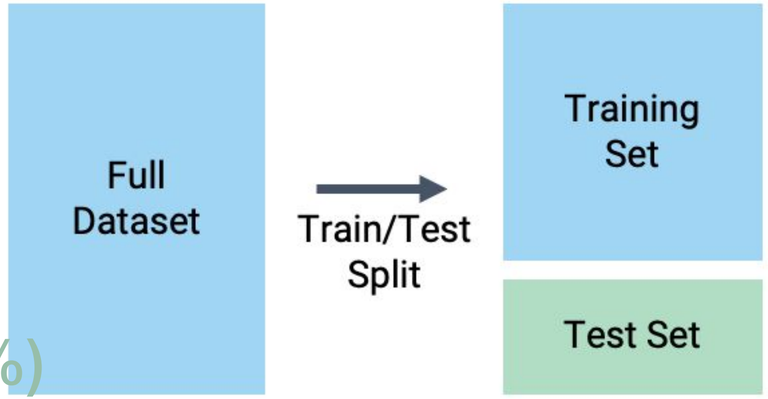
Most flexible





As model flexibility increases, train MSE will always decrease, but test MSE may not

Putting it all together



Supervised learning modeling workflow:

1. Split the data into a training (80%) and a test set (20%)
2. Train the model on the training data
3. Use the trained model to make predictions on the test data
4. Use a model evaluation metric to quantify the model's training and test performance
   - e.g. MSE for **regression** models
   - e.g. accuracy rate or error rate for **classification** models
5. Diagnose overfitting / underfitting by comparing the training and test error
6. Refine the model if needed!

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course code:
WSTAPX

You are using a multiple linear regression model to predict someone's height from their age and weight.

Which model evaluation metric would you use?

(A) Mean squared error

(B) Classification accuracy

(C) True positive rate

(D) False negative rate

(E) I'm not sure / I'm confused

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